

AJAY KASTURIRANGAN

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SUMMARY

Data Scientist with 4+ years of experience building data pipelines, AI tools, and analytics dashboards. Proficient in Python and GCP to engineer data workflows. Builds models using quantitative methods, time-series forecasting, and anomaly detection. Develops AI applications utilizing RAG, agentic modeling, and LLMs to automate research. Visualizes findings using Tableau and Power BI.

EXPERIENCE

Data Scientist / R&D Analyst, TIP Strategies

Aug 2023 – Present

- Built local RAG applications (Python, Ollama, LlamaIndex) to query 30 years of reports and summarize SEC filings, cutting research turnaround by 20%.
- Automated data pipelines via GCP and Python microservices and the Gemini API to generate dataset summaries and update client-facing Tableau dashboards, saving 3 hours per project and driving add-on revenue in 20% of engagements.
- Forecasted labor impacts using LSTM models and historical job data. Simulated expansion scenarios and reduced forecast variance by 40%, informing workforce planning for a \$17B regional investment.
- Deployed Hugging Face zero-shot classification models to categorize grant data and build knowledge graphs, replacing a rule-based system and improving accuracy by 30%.
- Built a geospatial transit framework (Python, NetworkX) to process GTFS data and score route capacity. Visualized networks in Tableau for 10+ projects, reducing analysis time from days to minutes.
- Deployed a local and custom OpenRouteService instance via Docker to process isochrone and drive-time queries, cutting processing time by 50% and enabling rapid site-selection analysis.
- Built a national freight data framework in Python to analyze supply chain networks and benchmark trade hubs, launching a new analytical product utilized in 3+ client projects.

Data Analyst, ICICI Lombard GIC

Feb 2021 – Aug 2022

- Integrated multi-terabyte datasets via Azure (Data Lake, Synapse) to migrate legacy SAP BO dashboards and develop 40+ Power BI financial reports, reducing report rendering time by 90% and centralizing operational tracking.
- Built topographic risk dashboards in Power BI using elevation and precipitation data to identify flood-prone infrastructure across 100+ miles of commercial highway projects, providing clients with realignment strategies to reduce insurance costs and support retention.

EDUCATION

New York University

Sep 2022 – Aug 2023

M.S. Applied Urban Science and Informatics

- Graduated with a 3.9 GPA and received the Scholastic Award.
- **Teaching Assistant:** Instructed students in supervised and unsupervised machine learning methods including ensemble methods and anomaly detection using Python, and supported a disaster management course utilizing Power BI for spatial and risk analysis.
- Conducted statistical analysis on eye-tracker data for an NYU Langone Health study, applying machine learning models to identify behavioral anomalies.

Manipal Institute of Technology

Jul 2017 – Jul 2021

B.Tech in Electronics Engineering

- **Minor in Data Science:** Built a strong quantitative foundation through applied projects and coursework in statistical inference, network analysis, linear algebra, calculus, and R programming.

SKILLS

- **Programming & Libraries:** Python (Pandas, Scikit-learn, NetworkX), SQL, R, DAX
- **Machine Learning & AI:** Generative AI (RAG, Agentic Modeling), Time-Series (LSTM), NLP (Hugging Face), Predictive Analytics, Anomaly Detection
- **LLM Stack & Frameworks:** LlamaIndex, LangChain, Ollama, Gemini API, Vector DBs (ChromaDB)
- **Cloud & Data Architecture:** GCP (BigQuery, Vertex AI), AWS (S3), REST API Integration
- **Visualization & Geospatial:** Tableau, Power BI, ArcGIS, OpenRouteService
- **Developer Tools:** Git, Docker